

Integration

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Integration

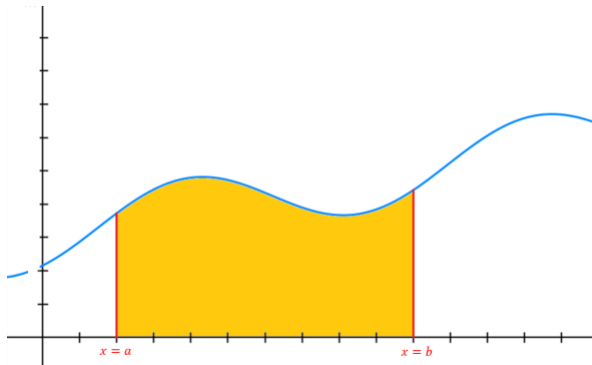


Figure 1: Graph of $y = f(x)$.

How to estimate area under the curve $y = f(x)$ and above the x -axis?

Integration

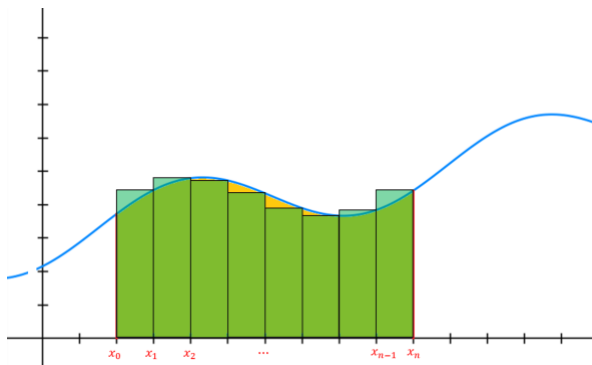


Figure 2: Graph of $y = f(x)$.

How to estimate area under the curve $y = f(x)$ and above the x -axis? One possible solution: sum over the rectangles.

Integration

Suppose $x_0 = a$, $x_n = b$ and subintervals $[x_{i-1}, x_i]$ for $1 \leq i \leq n$.
For each rectangle,

- ▶ the width is $\frac{b-a}{n}$,
- ▶ the length is $f(x_i)$ where $x_i = a + \frac{i}{n}(b-a)$,
- ▶ and thus the area of all rectangles is $\sum_{i=1}^n f(x_i) \cdot \frac{b-a}{n}$.

Note that we take the right endpoint x_i , and thus $f(x_i)$. One can also take the left endpoint x_{i-1} instead, and $f(x_{i-1})$ accordingly.

Integration

Theorem 1

Let f be a function that is continuous on a closed and bounded interval $[a, b]$. Then the following limit exists.

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \cdot \frac{b-a}{n}. \quad (1)$$

where $x_i = a + \frac{i}{n}(b-a)$ for $0 \leq i \leq n$.

Note that if $f \geq 0$ for all $x \in [a, b]$, then the limit above is the area bounded by the graph in Figure 1.

Integration

Following Theorem 1, the number of the limit in (1) is called *definite integral of f from a to b* , denoted by $\int_a^b f(x)dx$, i.e.,

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \cdot \frac{b-a}{n}. \quad (2)$$

From $\int_a^b f(x) \, dx$, f is called the *integrand*, a and b are called the *limits of integration*; lower and upper limits, respectively.

Example

Given $f(x) = x^2$, with $x \in [0, 1]$, we have $\frac{b-a}{n} = \frac{1}{n}$,
 $x_i = a + \frac{i}{n}(b-a) = 0 + \frac{i}{n}(1-0) = \frac{i}{n}$, and thus

$$\int_0^1 x^2 \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{i}{n} \right)^2 \cdot \frac{1}{n} \quad (3)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{i=1}^n i^2 \quad (4)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \quad (5)$$

$$= \lim_{n \rightarrow \infty} \frac{2n^3}{6n^3} \quad (6)$$

$$= \frac{1}{3}. \quad (7)$$

Note that (5) uses the sum of square rule, i.e.,

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

and (6) uses the leading term rule.

Exercise: find $\int_0^3 x^2 \, dx$.

Example

Given $f(x) = x^2$, with $x \in [0, 3]$, we have $\frac{b-a}{n} = \frac{3}{n}$,
 $x_i = a + \frac{i}{n}(b-a) = 0 + \frac{i}{n}(3-0) = \frac{3i}{n}$, and thus

$$\begin{aligned}\int_0^3 x^2 \, dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{3i}{n} \right)^2 \cdot \frac{3}{n} \\&= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{9i^2}{n^2} \cdot \frac{3}{n} \\&= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{27i^2}{n^3} \\&= \lim_{n \rightarrow \infty} \frac{27}{n^3} \sum_{i=1}^n i^2 \\&= \lim_{n \rightarrow \infty} \frac{27}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} \\&= \lim_{n \rightarrow \infty} \frac{54n^3}{6n^3} = 9.\end{aligned}$$

Rules for definite integrals

- ▶ $\int_a^b c \, dx = c \cdot (b - a).$
- ▶ $\int_a^b [f(x) + g(x)] \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx.$
- ▶ $\int_a^b \alpha f(x) \, dx = \alpha \int_a^b f(x) \, dx.$
- ▶ $\int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$

Indefinite integrals

Let f on (a, b) and suppose F is a function on (a, b) such that $F'(x) = f(x)$ for all $x \in (a, b)$. We say that F is an *antiderivative* for f on (a, b) .

For example, $f(x) = x^2$ and $F(x) = \frac{1}{3}x^3$. Then we have $F'(x) = f(x)$ for all $x \in \mathbb{R}$. Thus F is an antiderivative of f .

Integration

If F is an antiderivative for f on (a, b) , then all of the antiderivatives for f on (a, b) are of the form,

$$F(x) + C, \quad a < x < b, \quad (8)$$

where C is constant. Thus (8) represents a family of functions on (a, b) ; as there are infinitely many of them we call them *indefinite integral* of f . For example, the antiderivatives F for $f(x) = x^2$:

$$F(x) = \frac{1}{3}x^3, \quad F(x) = \frac{1}{3}x^3 + 1, \quad F(x) = \frac{1}{3}x^3 + 10, \\ F(x) = \frac{1}{3}x^3 - 1, \quad F(x) = \frac{1}{3}x^3 - 100$$

Integration

The indefinite integral of f is denoted by

$$\int f(x) \, dx, \quad (9)$$

that is,

$$\int f(x) \, dx = F(x) + C, \quad a < x < b, \quad (10)$$

where F is a function such that $F'(x) = f(x)$ for all $x \in (a, b)$ and C is an arbitrary constant. For example,

$$\int x^2 \, dx = \frac{1}{3}x^3 + C, \quad -\infty < x < \infty,$$

where C is an arbitrary constant.

Rules of indefinite integration

- ▶ $\int k \, dx = kx + C$
- ▶ $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$
- ▶ $\int k f(x) \, dx = k \int f(x) \, dx$
- ▶ $\int [f(x) + g(x)] \, dx = \int f(x) \, dx + \int g(x) \, dx$

Exercise 1

Perform the following integration.

1. $\int 2x^4 \, dx$

2. $\int (1 + x^3) \, dx$

3. $\int (x^3 - 2x + 4) \, dx$

4. $\int (2x - 3)(x^2 + 1) \, dx$

Apply the rules to evaluate definite integrals

$$\begin{aligned}\int_0^3 x^2 \, dx &= \left[\frac{x^3}{3} \right]_0^3 \\ &= \left(\frac{3^3}{3} \right) - \left(\frac{0^3}{3} \right) \\ &= 9.\end{aligned}$$

$$\begin{aligned}\int_{-1}^2 (x^2 - 2x + 3) \, dx &= \left[\frac{x^3}{3} - 2 \cdot \frac{x^2}{2} + 3x \right]_{-1}^2 \\ &= \left(\frac{8}{3} - 4 + 6 \right) - \left(\frac{-1}{3} - 1 - 3 \right) \\ &= 9.\end{aligned}$$

Exercise 2

Evaluate the following definite integrals.

1. $\int_0^3 2x^3 \, dx$

2. $\int_{-1}^2 (1 - 5x^4) \, dx$

3. $\int_{-2}^2 (x^4 - 3x^2 + 5) \, dx$

Trigonometry and Logarithmic Functions

Derivative and integration of Trigonometry and Logarithmic Functions:

$$\frac{d}{dx} \sin(x) = \cos(x)$$

$$\int \cos(x) \, dx = \sin(x) + C$$

$$\frac{d}{dx} \cos(x) = -\sin(x)$$

$$\int \sin(x) \, dx = -\cos(x) + C$$

$$\frac{d}{dx} \ln(x) = \frac{1}{x}$$

$$\int \frac{1}{x} \, dx = \ln(x) + C$$

$$\frac{d}{dx} e^x = e^x$$

$$\int e^x \, dx = e^x + C$$

References

S.K. Chung (2014), "Understanding Basic Calculus".
Available at <http://www.math.nagoya-u.ac.jp/~richard/teaching/f2015/BasicCalculus.pdf>